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NATIONAL UNIVERSITY - BALIWAG
ADVANCED MOBILE PROGRAMMING (AY 2026-2027)
LABORATORY ACTIVITY 2 SUBMISSION

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STUDENT NAME: Villaluna
PROJECT NAME: villaluna_advmobprog
ROOT WIDGET : VillalunaAdvMobProg
SUBMISSION : Villaluna_LabActivity2.pdf

1. PROJECT STRUCTURE & ARCHITECTURE

```
lib/  
├── main.dart  
├── constants/  
│   └── constants.dart  
├── models/  
│   └── product_model.dart  
├── providers/  
│   └── theme_provider.dart  
├── screens/  
│   ├── home_screen.dart  
│   ├── product_screen.dart  
│   ├── product_details_screen.dart  
│   └── settings_screen.dart  
├── services/  
│   └── product_service.dart  
└── widgets/  
    └── custom_text.dart  
  
assets/  
├── .env  
├── fonts/  
│   └── Poppins-Regular.ttf  
└── images/  
    ├── NU_shield.svg  
    ├── flutter.svg  
    └── nubdexchange_logo.png
```

2. REQUIRED FEATURES & ENHANCEMENTS IMPLEMENTED

- Enhancement 1: Product search bar
Location: lib/screens/product_screen.dart
Implementation: Case-insensitive search bar filtering product titles, categories, brands, and descriptions in real time.
 - Enhancement 2: Product details navigation
Location: lib/screens/product_screen.dart -> lib/screens/product_details_screen.dart
Implementation: Tapping a product card opens the scrollable details view showing thumbnail gallery, title, formatted price, rating, stock availability, brand, category, and full description.
 - Enhancement 3: Settings theme switch
Location: lib/screens/settings_screen.dart
Implementation: SwitchListTile interacting with Provider (ThemeProvider) to dynamically toggle between Light and Dark mode themes.
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3. TECHNICAL SUMMARY & COMPONENT RESPONSIBILITIES

- ProductModel: Converts API JSON into strongly-typed Dart objects with fallback null-safety handling.
- ProductService: Requests data over HTTP using flutter_dotenv HOST configuration, validates status code 200, and returns List<Product>.
- ProductScreen: Consumes ProductService via FutureBuilder and renders loading, error, empty, and success states in a responsive 2-column GridView.
- ThemeProvider: Manages theme state via ChangeNotifier and updates active Theme Data.
- Modular Pattern: Strict separation of concerns (models, services, providers, widgets, screens) for clean codebase maintainability.